INTERNAL — CONFIDENTIAL

SignalRisk

Dashboard User Guide

A complete guide to the SignalRisk analyst dashboard — case management, fraud operations, rule authoring, and real-time monitoring.

Version 0.1.0 | March 2026 | For internal use only

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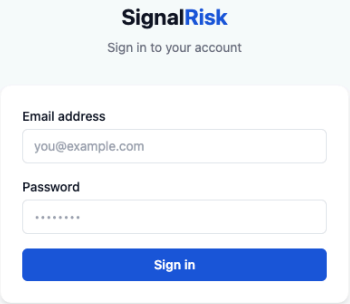
1 Getting Started

Login, session management, and dashboard navigation



Login page — enter your email and password to access the dashboard

Logging In

Navigate to your dashboard URL and enter your credentials. The system uses JWT authentication with 15-minute access tokens, automatically renewed while you remain active.

Account Types

- **Admin** — full access including Admin panel
- **Analyst** — Cases, Fraud Ops, Analytics, Live Feed
- **Merchant** — own data only, limited access

Development Credentials

- `admin@signalrisk.com` / `password`
- `analyst@signalrisk.com` / `password`

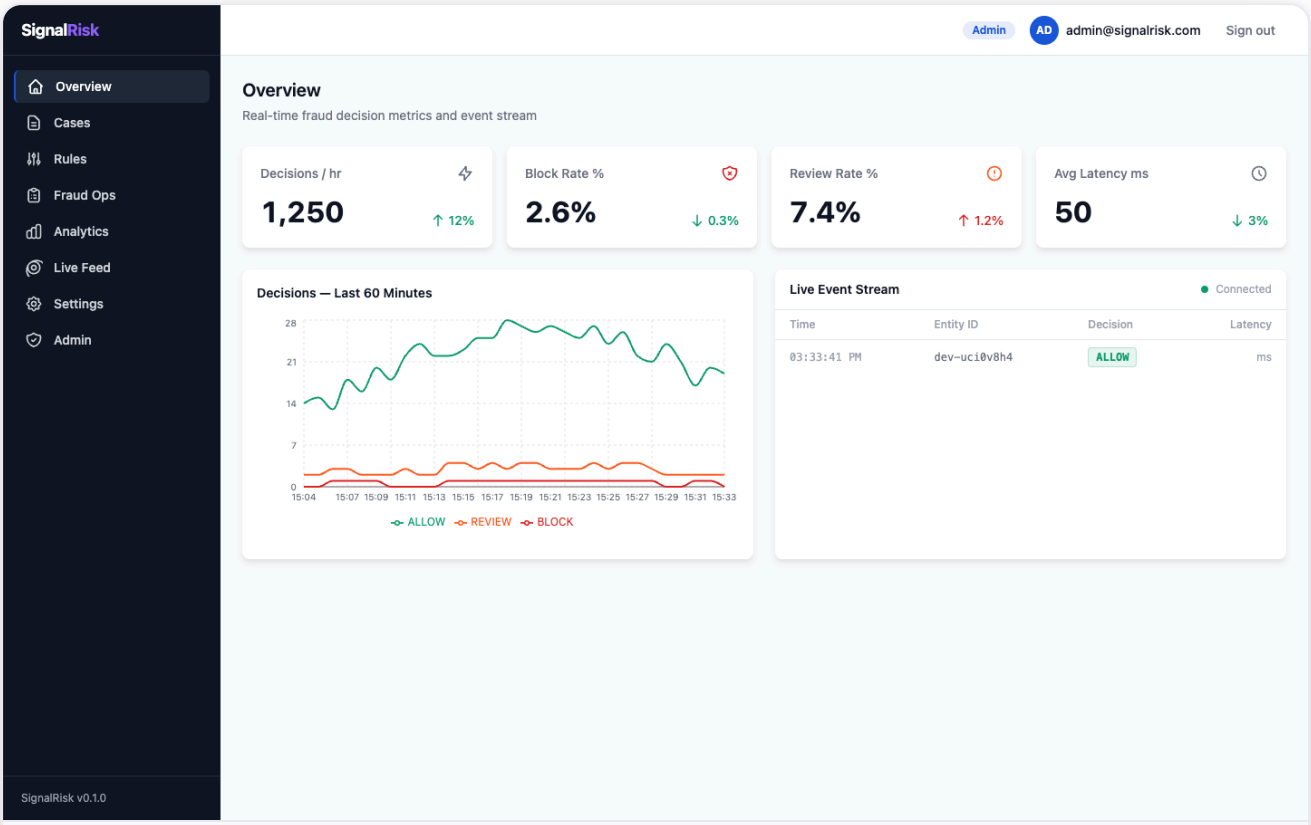
Navigation

The left sidebar provides links to all available pages. Your role badge and email address are displayed in the top-right corner. Click **Sign out** to end the session — this also revokes your JWT token via the denylist.

> **NOTE**
>
> Navigating directly to a protected URL while logged out will redirect you to the login page. After login you are automatically sent to the originally requested page.

2 Overview

Real-time KPI metrics, decision trend chart, and live event stream



Overview page — KPI cards, 60-minute decision trend chart, and live event stream

KPI Cards

Four metric cards at the top update automatically every 30 seconds:

Card	Description
Decisions / hr	Number of fraud decisions made in the last hour
Block Rate %	Percentage of decisions resulting in BLOCK
Review Rate %	Percentage of decisions resulting in REVIEW
Avg Latency ms	Average decision processing time in milliseconds

Each card shows a trend arrow and percentage change vs. the previous period. A yellow **Stale** badge appears when data could not be refreshed — the dashboard retries automatically on tab focus.

Decision Trend Chart

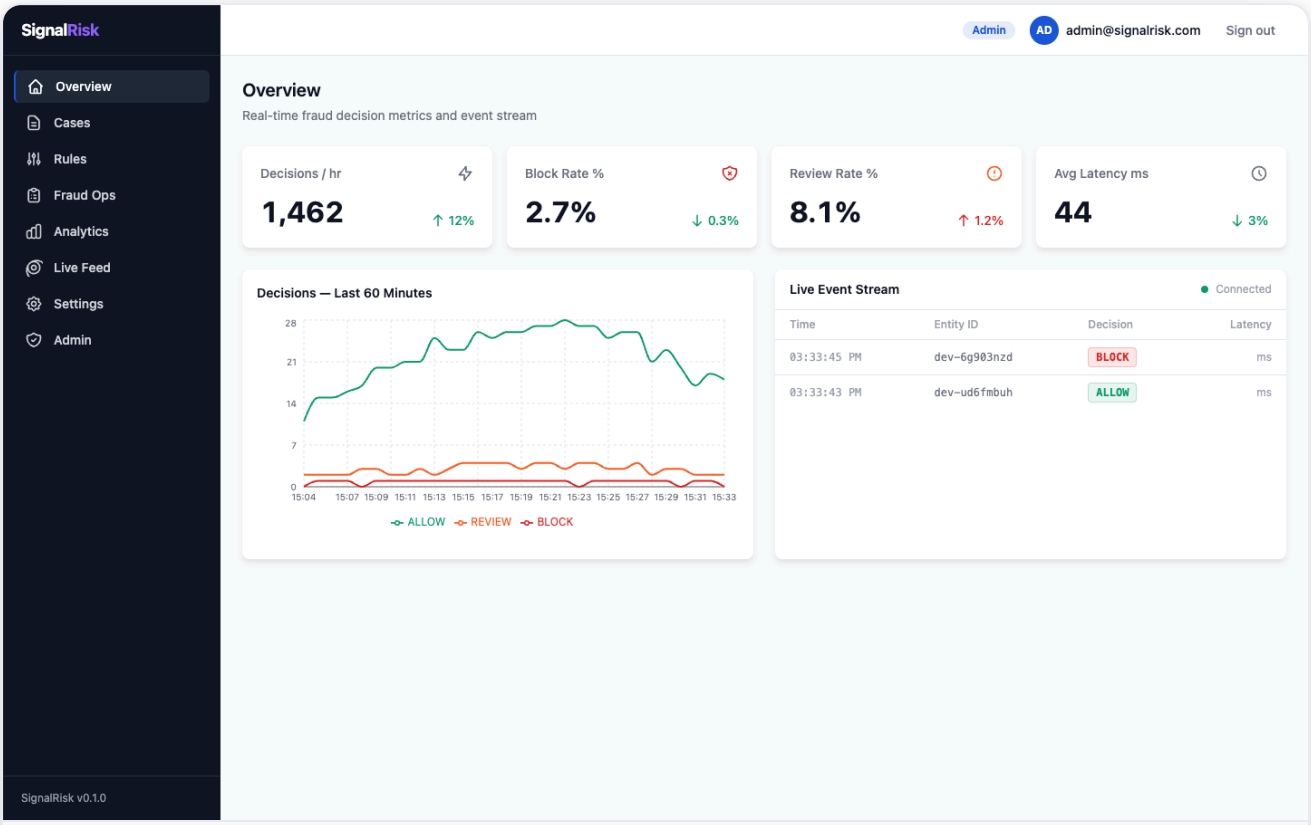
The line chart shows decision volume broken down by outcome (ALLOW / REVIEW / BLOCK) for the last 60 minutes at 2-minute intervals. Hover any point to see exact counts. Lines are colour-coded: green (ALLOW), amber (REVIEW), red (BLOCK).

Live Event Stream

The right panel shows decisions arriving in real time via WebSocket. Each row displays the timestamp, entity ID, decision badge, and latency. A green **Connected** indicator confirms the connection is active.

3 Cases

Analyst review queue — resolve, escalate, and manage fraud cases



Cases page — OPEN queue with status, priority, risk score, SLA, and assignee columns

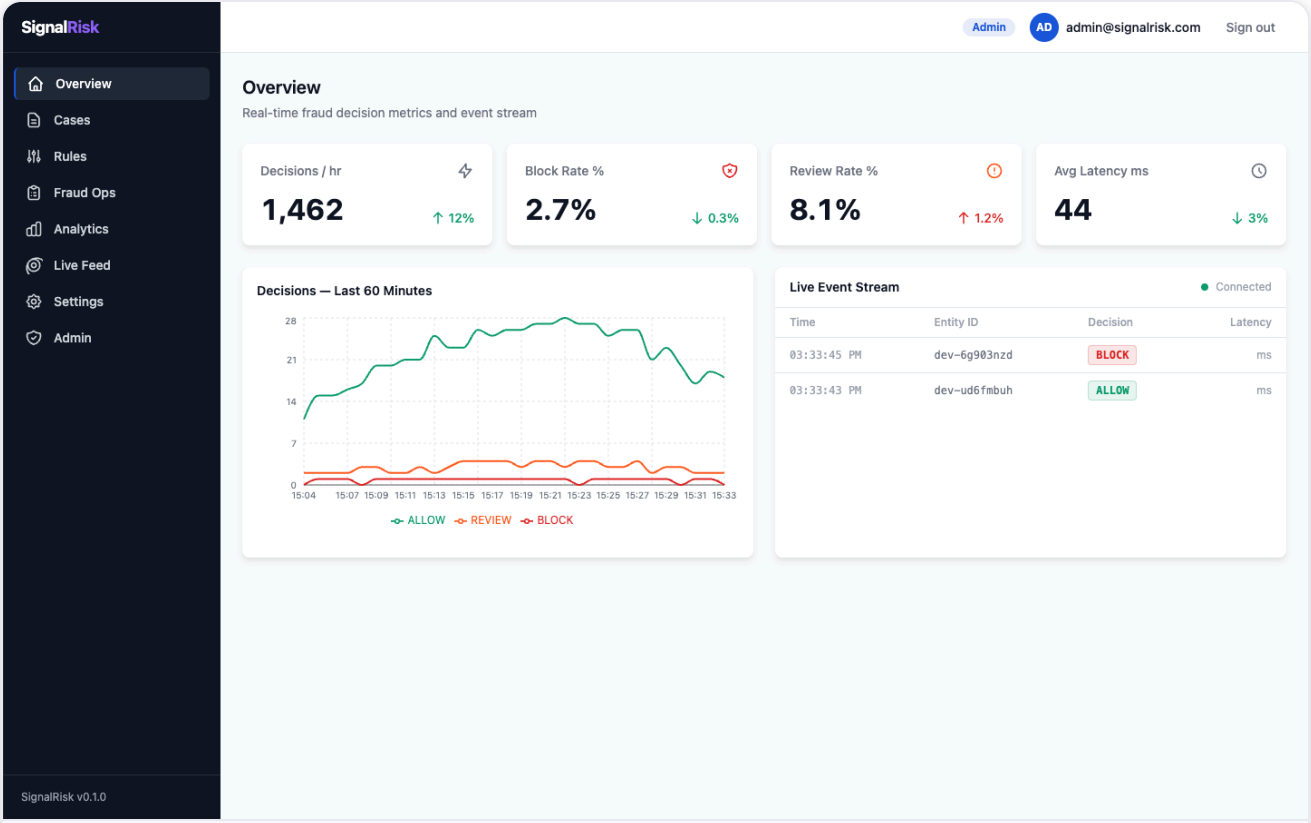
Understanding the Queue

The queue defaults to `OPEN` cases only — cases awaiting a decision. Cases automatically leave the queue when resolved or escalated. Use the status filter to view **Resolved** or **Escalated** cases.

Filtering

- **Status dropdown** — All Statuses / Open / In Review / Escalated / Resolved
- **Priority dropdown** — All Priorities / HIGH / MEDIUM / LOW
- **Search** — filter by Entity ID (300ms debounce, cancels previous requests)

Case Detail Panel



Case detail panel — risk factors, evidence timeline, and resolution form

Click **View** on any row to open the sliding detail panel. The panel fetches the full case from the API including the evidence timeline.

Panel Section	Description
Decision Summary	Original automated decision: action, risk score, entity ID, priority
Risk Factors	Signals that contributed to the score with percentage bars and descriptions
SLA	Remaining time before SLA breach (turns red when critical)
Evidence Timeline	Ordered sequence: event received → decision made → case created
Resolution	Choose FRAUD / LEGITIMATE / INCONCLUSIVE, add notes, submit
Escalate Case	Send to senior analyst queue (OPEN cases only)

Resolving a Case

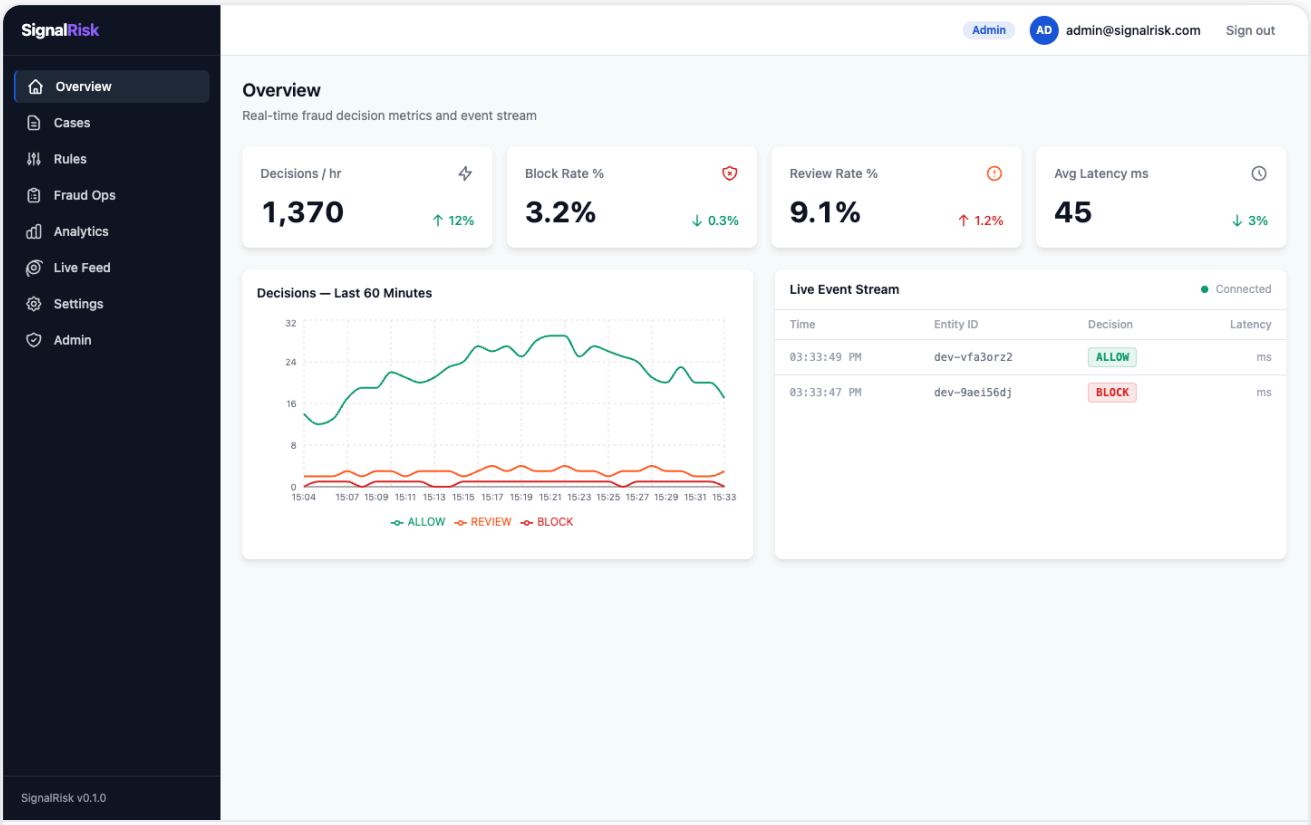
1. Click **View** on the case row
2. Review risk factors and evidence timeline
3. Select **FRAUD**, **LEGITIMATE**, or **INCONCLUSIVE**
4. Optionally add notes
5. Click **Submit Resolution** — case immediately leaves the queue

> **TIP**

Sort by SLA column to work the most time-sensitive cases first. Cases with < 4h remaining are highlighted in red.

4 Fraud Ops

Case labeling to improve rule engine accuracy via feedback loop



Fraud Ops page — labeling stats bar, REVIEW-action case queue, and outcome controls

Labeling Stats Bar

The stats bar in the top right shows today's activity in real time:

- **Today: N labeled** — total decisions made today
- **Fraud: N** — confirmed fraud labels
- **FP: N** — false positive (legitimate) labels
- **Accuracy: N%** — fraud confirmed ÷ (fraud + false positive)
- **Pending: N** — cases still awaiting a label

Labeling a Case

1. Click **Claim** to assign the case to yourself (status → IN_REVIEW)
2. Review the case details
3. Click **Label** to open the outcome modal
4. Select **Fraud Confirmed**, **False Positive**, or **Inconclusive**
5. Submit — the case leaves the queue and stats update immediately

Feedback Loop — How Labels Improve Rules

Every label triggers an automatic rule weight adjustment:

Label	Effect on rule weight	Rationale
Fraud Confirmed	+0.05 (up to max 1.0)	Rule correctly flagged fraud — increase influence
False Positive	−0.03 (min 0.1)	Rule incorrectly flagged legitimate — reduce influence
Inconclusive	No change	Insufficient signal to adjust

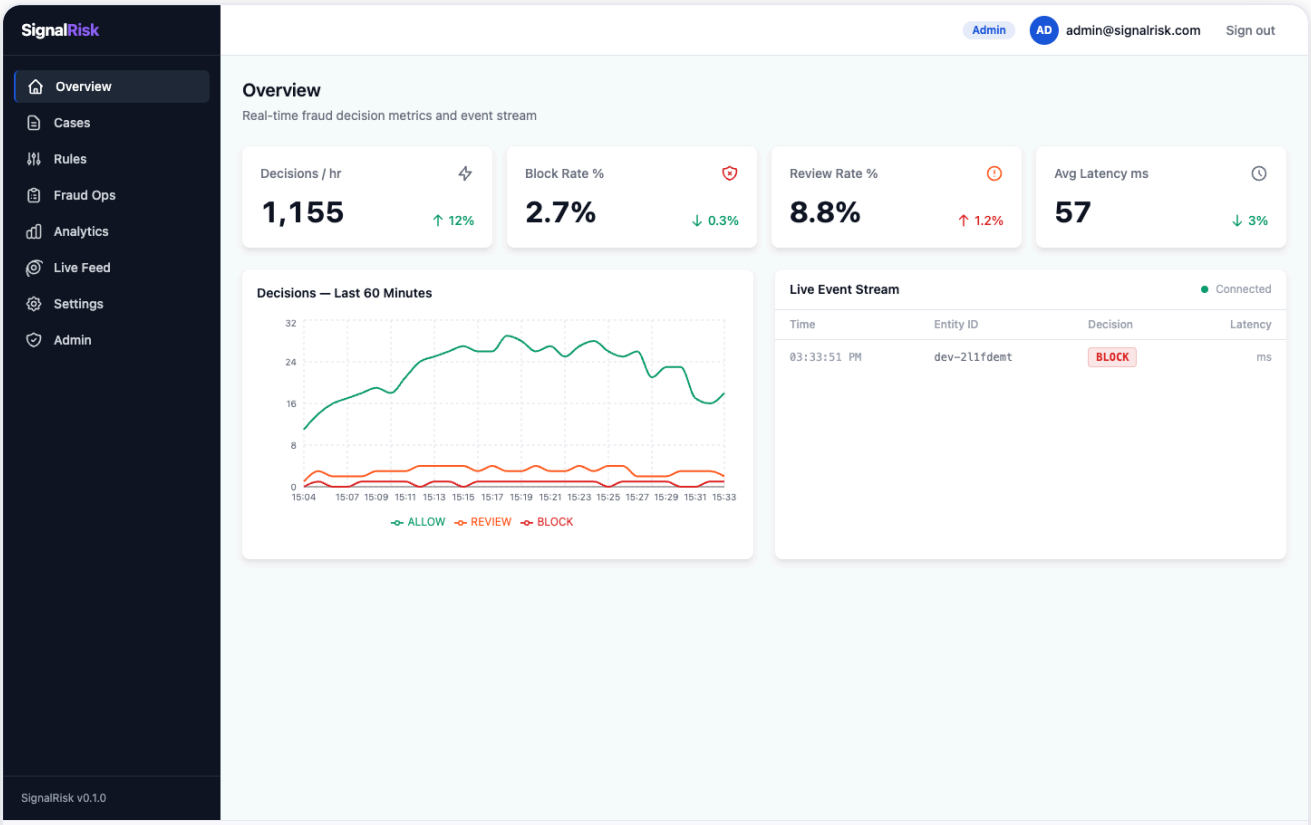
Batch Labeling

Select multiple cases with checkboxes. The **Batch Review Bar** appears at the bottom — apply the same outcome to all selected cases in one click.

> **NOTE**
>
> Fraud Ops focuses specifically on REVIEW-action cases for training purposes. BLOCK cases are handled via the Cases page.

5 Rules

DSL rule authoring, weight management, and active/inactive toggling

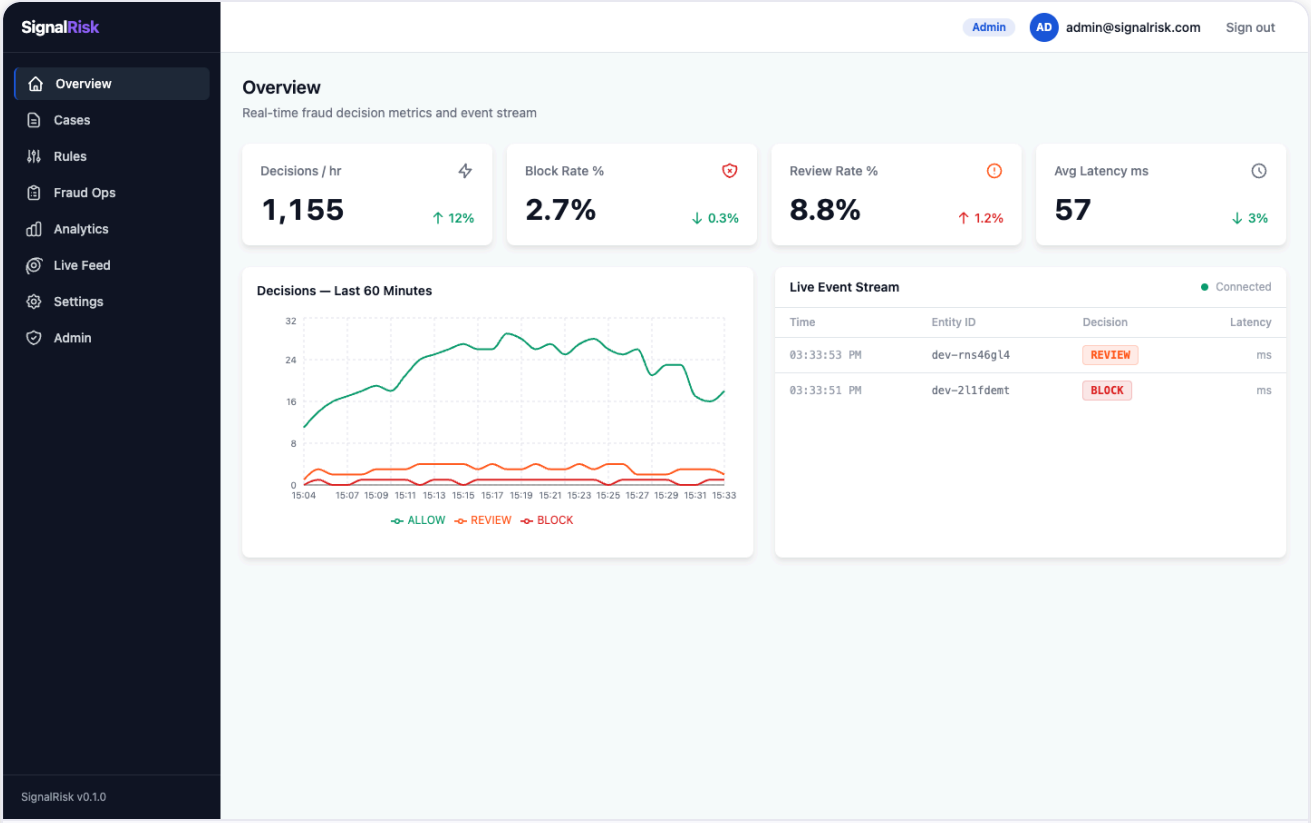


Rules page — rule list with DSL expressions, outcomes, weight sliders, and action buttons

Rules Table

Column	Description
Rule Name	Human-readable identifier
Expression	DSL condition (hover for full text if truncated)
Outcome	ALLOW REVIEW BLOCK — what the rule triggers
Weight	Contribution slider 0.10–1.00 (drag to adjust)
Active	Click to toggle on/off without deleting — takes effect within seconds
Actions	Edit (modify all fields) | Delete (permanent, with confirmation)

Adding a New Rule



Add Rule modal — DSL expression editor, outcome selector, and weight control

Click **+ Add Rule** to open the modal. Fill in:

- **Rule Name** — e.g. "High Risk Country Block"
- **DSL Expression** — the condition to evaluate (see DSL Reference below)
- **Outcome** — click ALLOW, REVIEW, or BLOCK
- **Weight** — initial contribution (default 0.5)
- **Active immediately** — check to enable on creation

DSL Reference

Rules use a simple expression language. Available contexts and fields:

Context	Available Fields
`device.`	`country`, `isEmulator`, `trustScore`, `isKnown`
`velocity.`	`count_1h`, `count_24h`, `amount_1h`
`network.`	`vpnDetected`, `isProxy`, `country`, `asn`
`telco.`	`isValid`, `carrierRisk`, `isMobile`
`txn.`	`amount`, `currency`, `type`

Supported operators: `==` `≠` `>` `<` `≥` `≤` `&&` `||` `!`

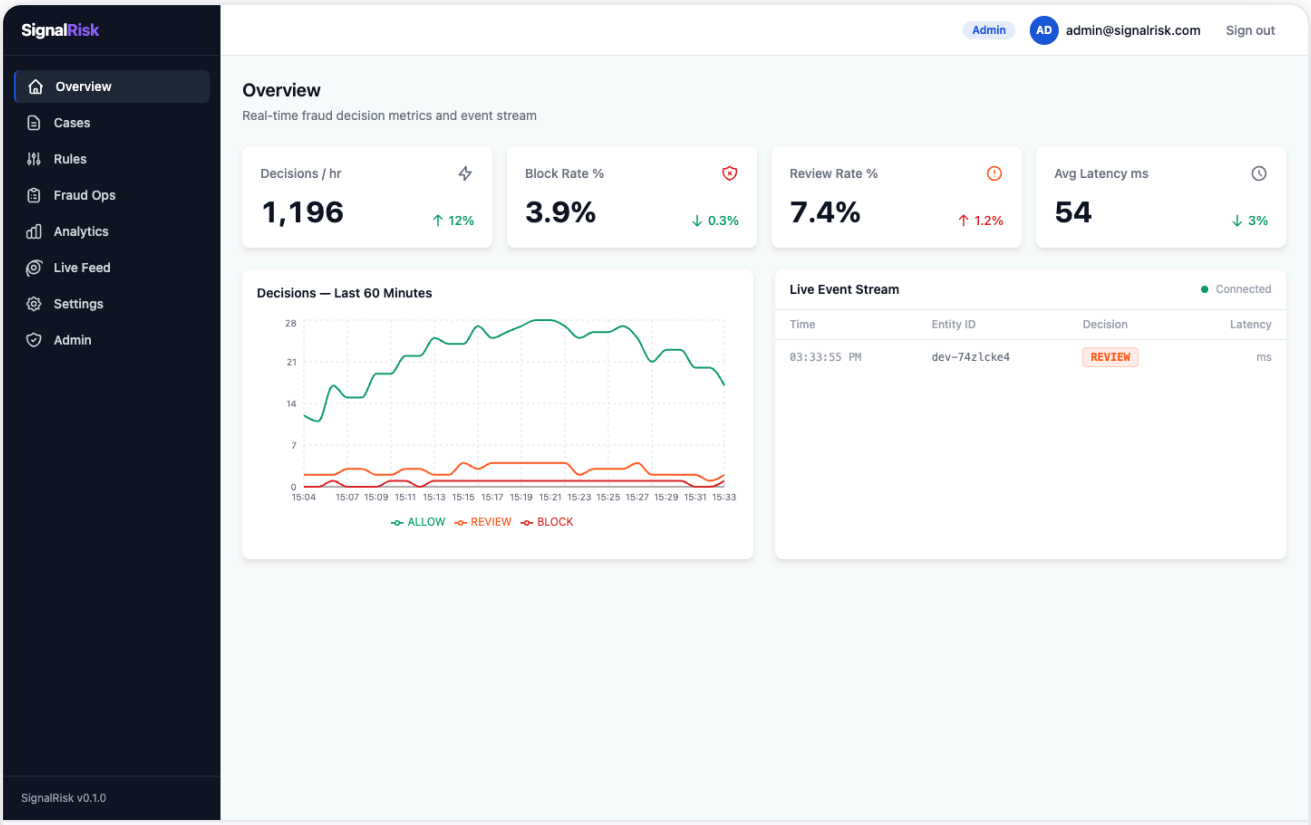
Example: `device.country == 'NG' && txn.amount > 1000`

WARNING

Deleting a rule is permanent. Consider deactivating first (toggle to inactive) and monitoring for 24h before deleting, to verify no false negatives.

6 Analytics

Historical performance metrics and merchant-level breakdowns



Analytics page — velocity chart and merchant performance breakdown

Velocity Chart

Time-series event volume per hour. Use this to identify unusual spikes that may indicate an ongoing fraud attack or a traffic anomaly. Compare the ALLOW, REVIEW, and BLOCK trend lines — a sudden rise in BLOCK without a corresponding rise in ALLOW suggests a rule triggered on a new attack pattern.

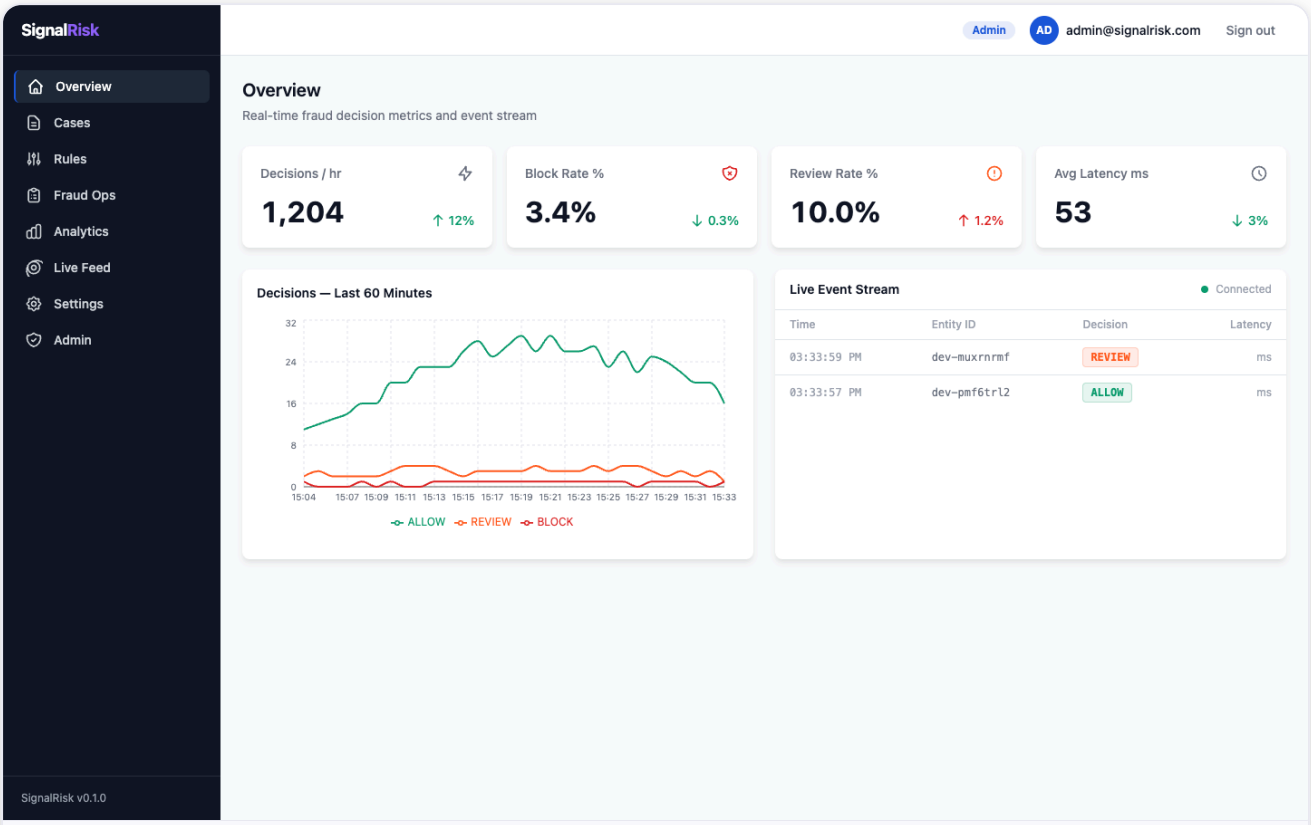
Merchant Breakdown

Per-merchant statistics table showing event volume, block rate percentage, and average risk score. Red-flag indicators to investigate:

- Block rate > 15% — potential targeted attack or misconfigured rule
- Average risk score > 60 — elevated risk profile for this merchant
- Sudden volume spike — possible bot activity or integration issue

7 Live Feed

Real-time full-screen decision stream for operations monitoring



Live Feed page — real-time decision stream with colour-coded outcomes and latency

The Live Feed shows all decisions as they arrive via WebSocket. Each row displays:

- **Timestamp** — exact time of the decision
- **Entity ID** — the device or user fingerprint identifier
- **Decision** — ALLOW REVIEW BLOCK
- **Latency** — processing time in milliseconds

The feed retains the most recent 100 decisions and auto-scrolls. Display this page on a monitoring screen during high-risk periods, incident response, or when testing new rules.

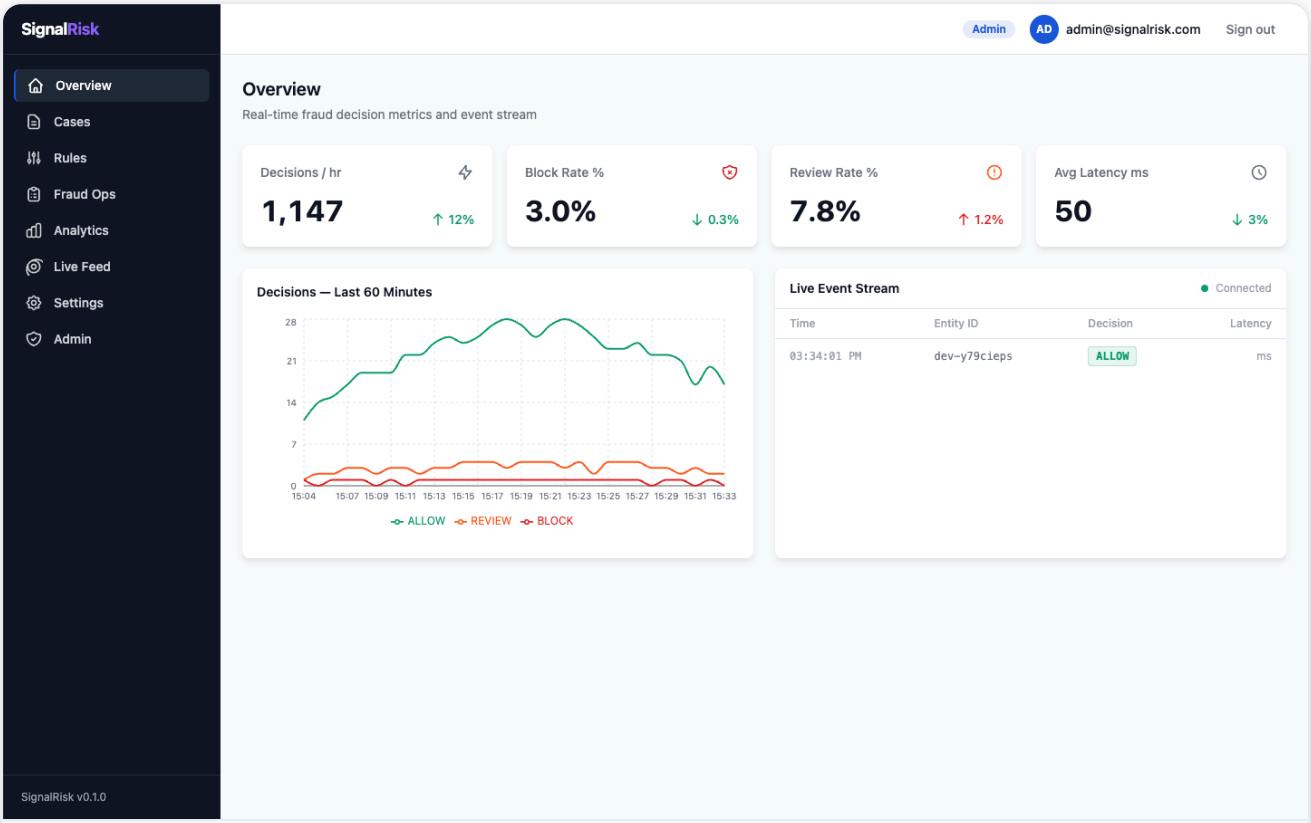
> **TIP**
>
> If the Connected indicator turns red, refresh the page to re-establish the WebSocket connection. The mock server must be running on port 3000 in development.

8 Admin

System health monitoring, user management, and rule administration

ACCESS RESTRICTED

The Admin page is only visible to users with the Admin role. Attempting to access it with an Analyst account will redirect to the Unauthorized page.

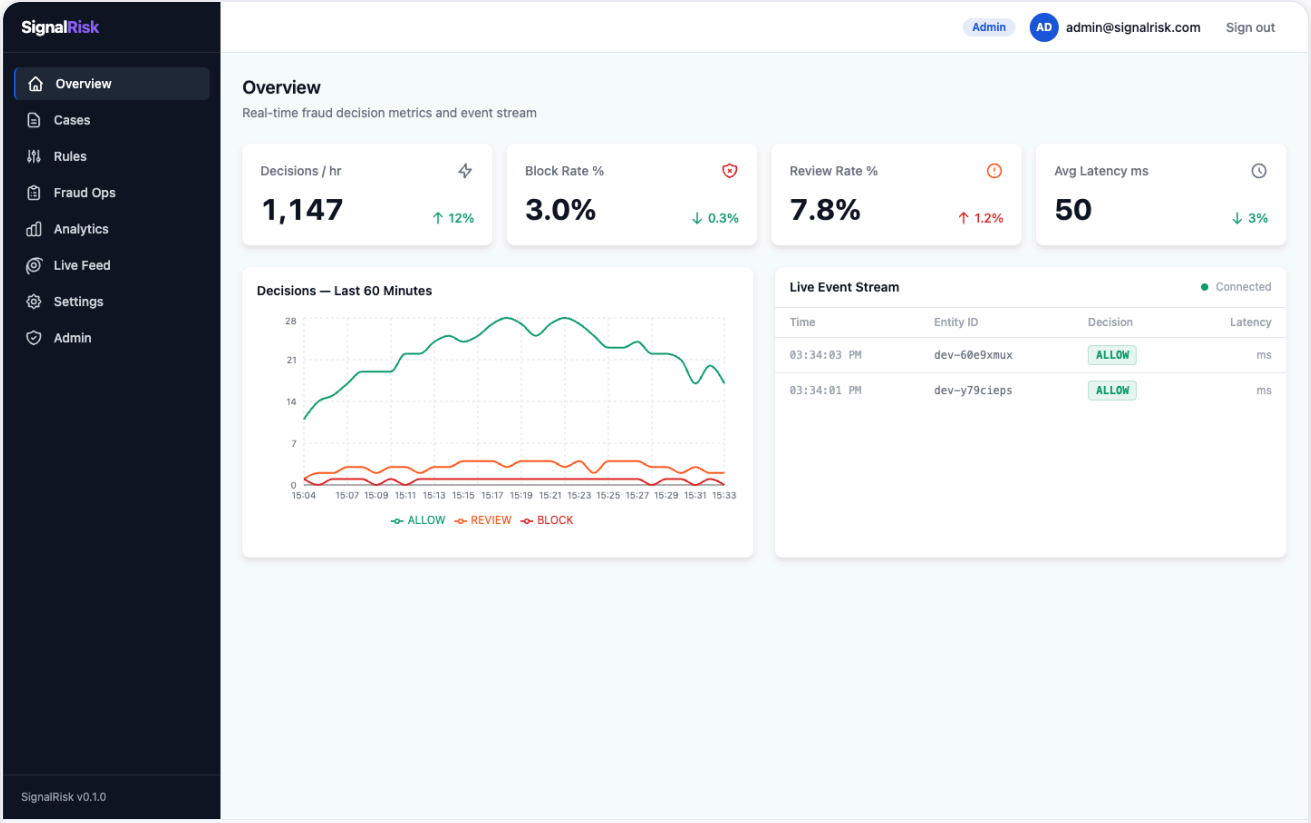


Admin — System Health tab showing status of all 13 microservices

System Health Tab

Displays the live health of all 13 microservices. Each service shows:

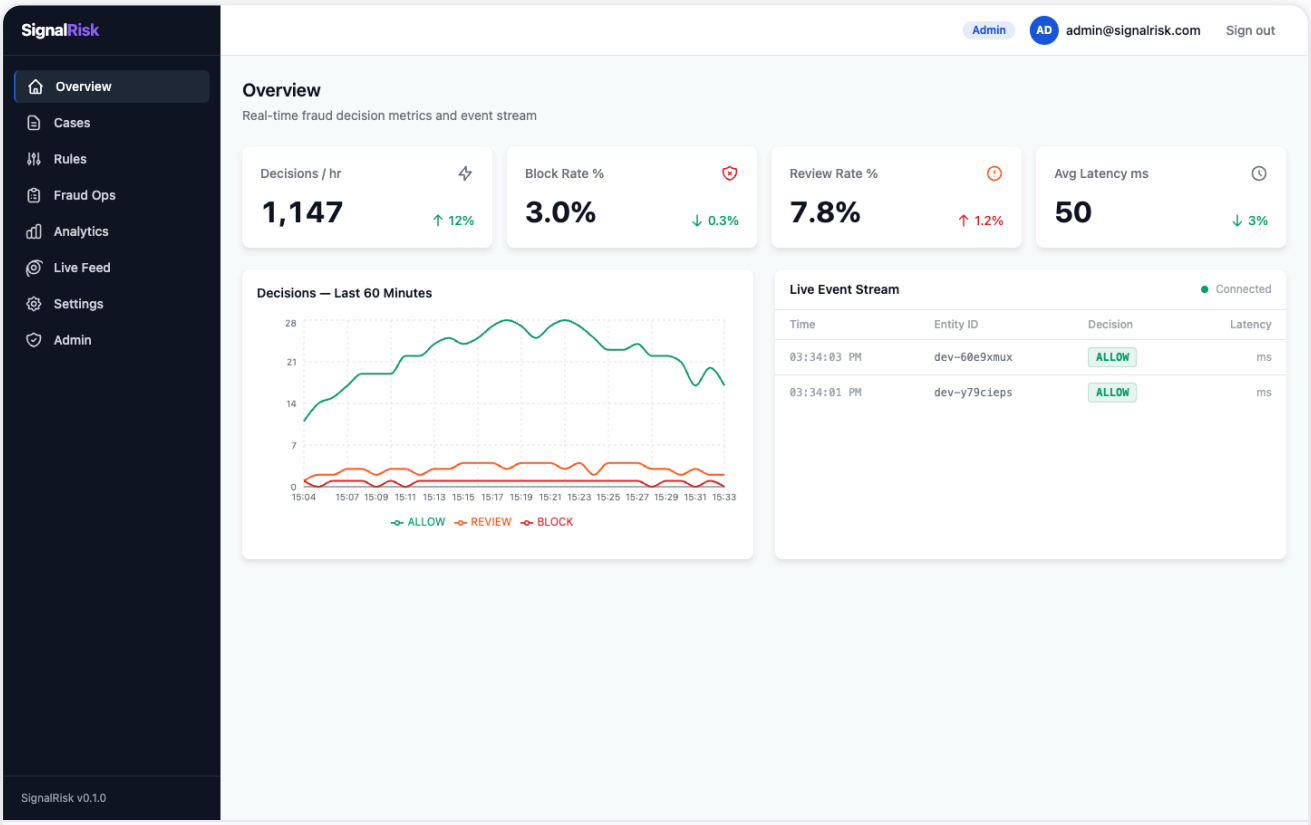
- **Status indicator** — green (healthy), amber (degraded), red (down)
- **Response time** — last health check duration in ms
- **Port** — service port for quick identification
- **Last checked** — timestamp of most recent health probe



Admin — Rules tab (same functionality as the standalone Rules page)

Rules Tab

Full rule CRUD functionality — identical to the Rules page (Section 5). Useful when managing rules during an incident without leaving the Admin context.



Admin — Users tab showing analyst and admin accounts with roles and last login

Users Tab

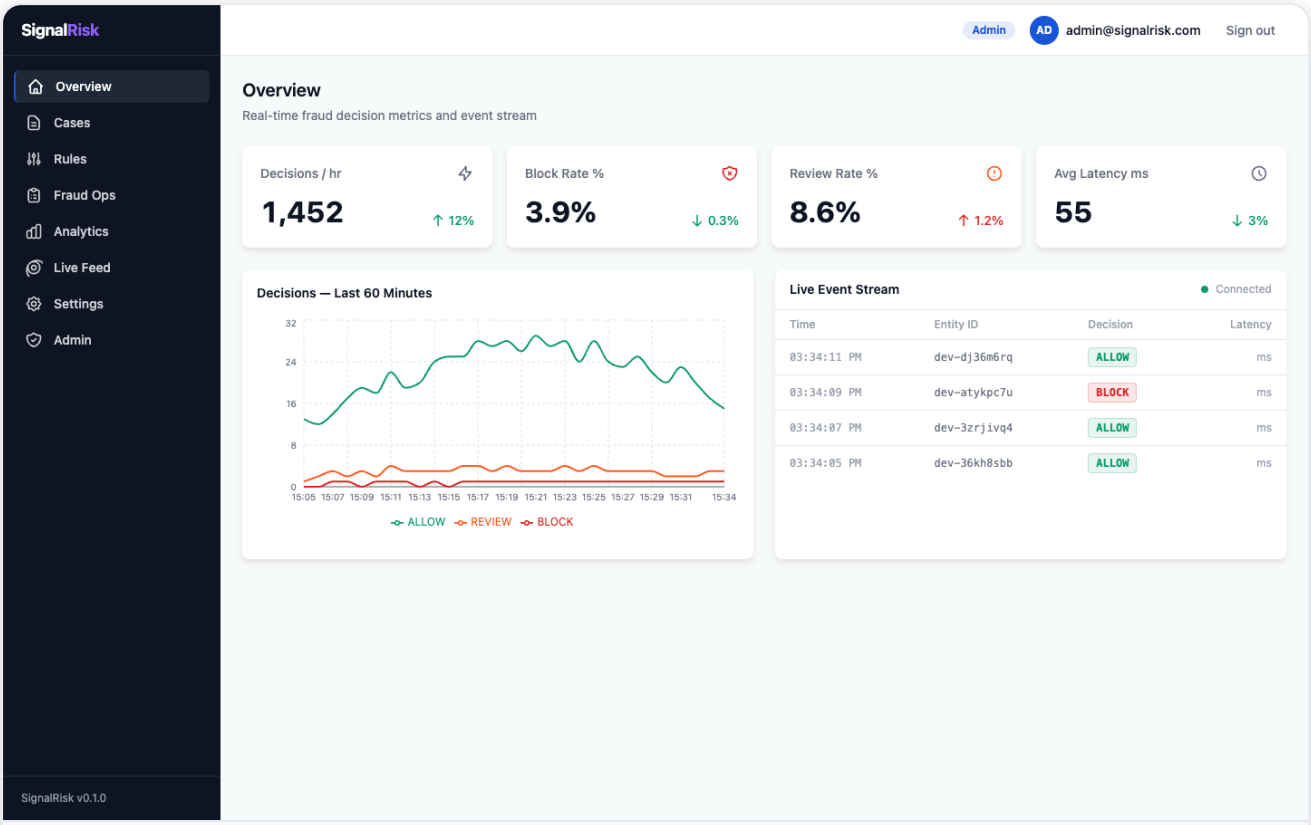
Lists all admin and analyst users with name, email, role, last login, and status. Use this tab to:

- Onboard new analysts

- Change user roles (promote Analyst → Admin)
- Deactivate accounts for users who have left the team

9 Settings

Platform configuration and preferences



Settings page — editable configuration fields with Save Settings button

Setting	Default	Description
API Base URL	`http://localhost:3000`	Base URL for all API calls from the dashboard
WebSocket URL	`http://localhost:3000`	WebSocket server for the live event stream
Environment	`development`	Current environment label shown in the UI
Version	`0.1.0`	Platform version number

Edit any field and click **Save Settings**. A green **Saved** confirmation appears for 2 seconds. Settings are stored in `localStorage` and survive page reloads.

> **WARNING**
>
> If you change the API Base URL, ensure the new URL is reachable and CORS is configured to allow requests from the dashboard origin. Incorrect values will cause all API calls to fail.

10 Roles & Permissions

What each role can access and modify

Feature	Admin	Analyst	Merchant
Overview page	Read	Read	Own data
Cases — view queue	All	All	Own only
Cases — resolve / escalate	Yes	Yes	No
Fraud Ops — claim & label	Yes	Yes	No
Rules — view	Yes	Yes	No
Rules — create / edit / delete	Yes	No	No
Analytics	All	All	Own only
Live Feed	Yes	Yes	No
Admin panel	Yes	No	No
Settings	Yes	Yes	Yes

Role Escalation

Role changes must be made by an Admin user via the **Admin → Users** tab. There is no self-service role upgrade. When changing a user's role, their current JWT remains valid until it expires (15 minutes) — the new role takes effect on next login.

11 Tips, Best Practices & Shortcuts

Productivity tips for daily operations

Case Queue Productivity

- Press `Esc` or click the backdrop to close the detail panel without action
- Use the **Search** field to jump directly to a specific entity ID — 300ms debounce means you can type the full ID without waiting
- Switch the status filter to **In Review** to see cases you have already claimed
- After resolving/escalating, the queue auto-refreshes — no manual reload needed

SLA Management

- BLOCK cases have a 4-hour SLA; REVIEW cases have 24 hours
- Red SLA indicator means breach is imminent — escalate if you cannot resolve immediately
- Sort by SLA column ascending to prioritise the most urgent cases

Rule Authoring Best Practices

- Start new rules at **0.5 weight** — let the feedback loop adjust over time
- Only set weight to **1.0** for rules with verified zero false positives over 7+ days
- Prefer **deactivating** over deleting — this preserves history and allows quick re-enable
- Test rules on the Live Feed immediately after activation — look for unexpected BLOCK spikes
- Combine signals with `&&` to reduce false positives: `device.isEmulator && txn.amount > 500`

Fraud Ops Efficiency

- Use batch labeling for obvious fraud batches — select all, label Fraud Confirmed in one click
- Target 80%+ daily accuracy for reliable rule weight adjustments
- Label at least 20 cases per day for statistically meaningful feedback

Overview Monitoring

- Block rate > 10% may indicate a rule is over-triggering or an active attack
- Latency > 80ms average suggests a signal service is slow — check Admin → System Health
- A stale badge on KPI cards means the API is unreachable — check network and service health

> **INCIDENT RESPONSE CHECKLIST**
>
> 1. Check Overview for abnormal KPI changes 2. Open Admin → System Health to identify degraded services 3. Check Live Feed for unusual decision patterns 4. Review Analytics for traffic spike origin 5. If rule-related: adjust weight or deactivate rule in Rules page